

# CLOUD SECURITY FRAMEWORK

## Contents

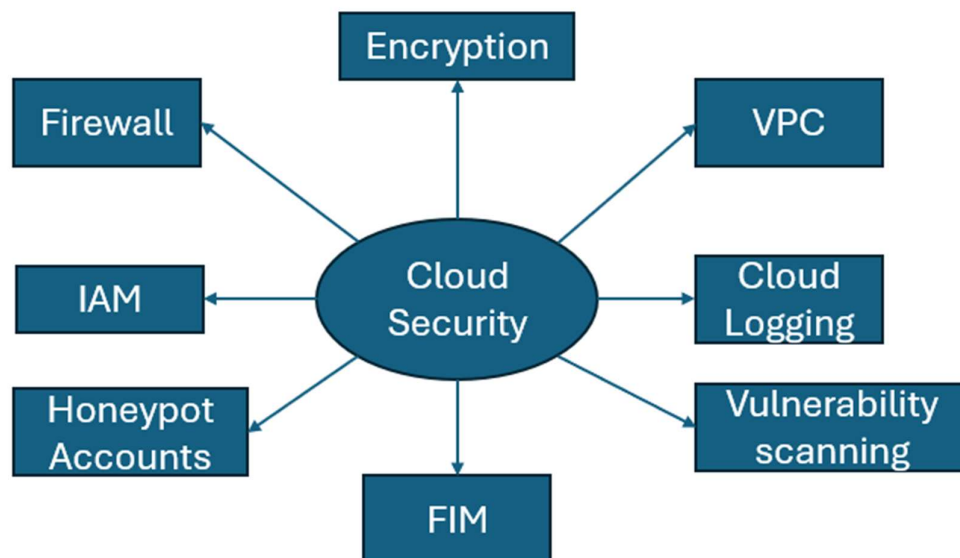
**Introduction** ..... 2

**Basic Cloud Security Protections**..... 3

## Introduction

This cloud security framework is designed to help companies and individuals secure cloud servers. This cloud security framework was developed as currently cloud servers are used in some of the biggest applications such as cloud storage, AI/ML and database hosting. This means that if an attacker were able to exploit a cloud server, they could access sensitive user data or disrupt business operations. This cloud security framework will explain security features in the most understanding way possible.

This cloud security framework cannot promise perfect security for cloud servers. However, it can promise to provide more security and raise the cost needed for an attacker to breach the cloud infrastructure. This framework is mainly targeted for companies with cloud servers and rarely individuals using self-hosted cloud servers. Make sure to adapt the protections mentioned in this framework to security resources and compliance standards.



This spider diagram shows some of the main cloud protections that can be implemented to protect cloud servers. For those who may not know, VPC stands for virtual private cloud. It is used to keep cloud resources on a private network and control traffic between services. IAM stands for identity and access management. It is used to verify identity and control cloud access. FIM stands for file integrity monitoring. FIM ensures that the files stored in the server haven't been corrupted or tampered with. You can find more information on protections in the sections below.

## **Basic Cloud Security Protections**

- Every cloud server should have these 5 basic protections.
- There are 5 main protections that can be implemented.
- The first protection is IAM.
- IAM also known as identity and access management ensures that a cloud server verifies user's identity and controls access to cloud resources.
- Identity verification includes usernames, passwords and MFA via authentication app or other methods to verify a user's identity.
- Access control includes giving each user a role which allows them to access certain cloud resources.
- This is important because it ensures that an attacker cannot easily access cloud resources even if they compromised an account.
- The second protection is encryption.
- Encryption is used to protect the data stored at rest on the server and during transit during client-server interactions.
- For data at rest, I would recommend using AES-256 and store the key either using HSM (Hardware Security module) or KMS (Key management service).
- For data in transit, I would recommend using TLS encryption within HTTPS to ensure that an observer cannot read the data.
- This is important because it ensures that the attacker cannot read the data while it is stored at rest or intercept it while it is in transit.
- The third protection is FIM.
- FIM also known as file integrity monitoring ensures that the data stored in the server hasn't been corrupted or tampered with.
- FIM can be implemented using tools such as Tripwire, AIDE, OSSEC or cloud-provider FIM.
- This is important because it ensures that the integrity of data is monitored and allows defenders to actively respond to tampering attempts.
- The fourth protection is anti-virus software.
- Anti-virus software is a software application that handles malware on a endpoint device like laptop or in this case server.
- Anti-virus software can be implemented by using built-in options such as Windows Defender or ClamAV or third-party options such as Bitdefender GravityZone.
- This is important as malware especially ransomware is one of the most dangerous threats to cloud infrastructure.
- The fifth and final protection is firewall.
- A firewall is a software application designed to filter out malicious traffic from usual traffic.
- I would recommend configuring the firewall to block vulnerable ports and block certain programs and services.

## **Advanced Cloud Security Protections**

- Cloud servers can implement more advanced protections that may be used against higher threat models such as organised cybercriminals.
- There are 5 main advanced protections that can be implemented.
- The first protection is VPC.
- VPC also known as virtual private cloud isolates cloud resources on a private network
- VPCs can be implemented through cloud console, CLIs, IaC (Infrastructure as code) and SDKs
- This is important because they ensure that all cloud resources cannot be accessed or used by everyone.
- The second protection is private subnets.
- A private subnet is a subnet within a VPC that has no direct route to the public internet.
- A private subnet can be implemented by defining it during a setup of an VPC through the provider's tool.
- This is important because it ensures that sensitive data is not exposed to the public internet.
- The third protection is vulnerability scanning and automated patching.
- Vulnerability scanning is using an automated tool configured to find known vulnerabilities within the cloud server's configurations.
- Automated patching is using an automated tool to patch and fix vulnerabilities that were found by the automated vulnerability scanning.
- This is important because it ensures that vulnerabilities are found and patched without human intervention in most cases.
- The fourth protection is honeypot accounts.
- Honeypot accounts are fake accounts created for the purpose for tricking and detecting attackers without compromising the real accounts.
- Honeypot accounts can be implemented by setting up fake credentials and access permissions within a network that are set to alert an admin once used.
- This is important because it ensures that attacker can be detected while they think they have gained a foothold into the system.
- The fifth protection is IP whitelisting.
- IP whitelisting is where a firewall on a cloud server is configured to only communicate or accept traffic from certain IP ranges.
- This can be implemented by configuring the firewall to only allow traffic from a trusted IP range on the public internet or privately.
- This is important because it prevents an attacker from sending malicious traffic to a cloud server that doesn't allow traffic from the attacker's IP address.
- There are other advanced protections such as session timeouts, SIEM alerts, security groups and snapshot scheduling.

## **Cloud Security Framework Trade-Offs**

- **This cloud security framework has trade-offs just like any other system I have created.**
- **There may also be further implementation challenges that users or experts may face.**
- **I have identified 5 main trade-offs.**
- **The first trade-off is cost.**
- **This trade-off exists because cloud server infrastructure can be extremely expensive to set up and maintain.**
- **This means that some companies may be able to barely afford cloud servers and implement the security protections needed.**
- **This trade-off can be managed by allocating a specific budget, using pre-existing infrastructure and minimal setup where possible.**
- **The second trade-off is time-consumption.**
- **This trade-off exists as it can take long time to configure a cloud server especially for the purpose you need for.**
- **This means that some IT teams or people may not have the time to focus on cybersecurity due to busy schedules.**
- **This trade-off can be managed by using time-blocking, specific and measurable goals and a priority system for allocating time for configurations.**
- **The third trade-off is complexity.**
- **This trade-off exists because advanced security features such as VPC and Private subnetting can be difficult to configure especially without technical expertise.**
- **This means that individuals or companies without an IT team or individual**
- **This trade-off can be managed by using structured learning processes, hiring a technical expert or automating the process where possible.**
- **The fourth trade-off is cognitive load.**
- **This trade-off exists as it can be mentally exhaustive to configure and maintain a cloud server especially a secure one.**
- **This means that individuals or companies may exhaust themselves and face burnout attempting to configure cloud servers.**
- **This trade-off can be managed by ensuring breaks are built in the time management, automated processes and having a team if possible.**
- **The fifth and final trade-off is compatibility.**
- **This trade-off exists because security features configuration is dependent on the provider's tools and the server's purpose.**
- **This means that some companies or users may not be able to implement all protections**
- **This trade-off can be managed by performing research on provider tools before purchase and accepting that some security features won't be supported.**